THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** a feather	Air resistance matters for feathers.
2	**(d)** 20 per cent	30/150.
3	**(d)** 10 g	Conservation of mass.
4	**(b)** 16	Sum of electrons in a neutral atom.
5	**(b)** 12 m/s	v = at.
6	**(b)** 6 kg	m = F/a.
7	**(c)** zero	cos 90 = 0.
8	**(b)** 0.68 m	v/f.
9	**(a)** hypertonic	Water leaves.
10	**(b)** increase in girth	Secondary growth.
11	**(b)** Protista	Euglena.
12	**(d)** vegetative propagation by suckers	Banana is seedless.
13	**(c)** 500 s	d/v.
14	**(a)** slowly stops swinging	Friction at the pivot and air resistance.
15	**(c)** fractional distillation	Boiling-point difference.
16	**(d)** $Al_2(SO_4)_3$	Al3+ and SO4 2-.
17	**(b)** 12	2, 8, 2.
18	**(d)** 5 m/s	100/20.
19	**(d)** 200 N	600/3.
20	**(d)** 9	0.06 x 150.
21	**(b)** greater for the 256 Hz fork	lambda proportional to 1/f.
22	**(b)** 8	Same number.
23	**(d)** squamous epithelium	Alveoli.
24	**(b)** gills throughout life	Fish breathe with gills.
25	**(a)** bisexual flower	Has both.
26	**(b)** coastal low-lying areas	Flooding.
27	**(b)** 2500	1 g/cm3 = 1000 kg/m3.
28	**(c)** sublimation, dissolving and filtering, evaporation	Camphor first.
29	**(c)** Mg and Cl	Metal and non-metal.
30	**(d)** 35.5 u	26.25 + 9.25.
31	**(a)** 30 m/s	t up = 3 s; u = gt.
32	**(b)** 12 N	Equal to the applied force at the verge of motion.
33	**(a)** 4	Effort arm/load arm.
34	**(a)** quality or timbre	Waveform.
35	**(d)** mitochondria	Energy.
36	**(b)** hinge joint	One plane.
37	**(a)** Animalia, Chordata, Mammalia, Carnivora, Felidae, Panthera, tigris	Hierarchy.
38	**(b)** 46	23 + 23.
39	**(d)** evaporation	Heating.
40	**(b)** 8 m/s	100/12.5.

41	**(b)** 10 g	4/100 x 250.
42	**(c)** 28 g	0.8235 x 34.
43	**(a)** 1	2, 8, 8, 1.
44	**(b)** 100 m	s = (1/2)at squared.
45	**(a)** 6 N	Change in momentum/time.
46	**(a)** 20 m/s	sqrt(2gh).
47	**(c)** 20	3.4/0.17.
48	**(a)** growing region	Meristem.
49	**(d)** vascular cambium	Rings.
50	**(c)** monocot	Monocot.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** absent	Pregnancy.
2	**(a)** nitrogen fixation	Bacteria and lightning.
3	**(d)** supports the law	Agreement supports it.
4	**(c)** sublimation	Iodine sublimes.
5	**(c)** $(NH_4)_3PO_4$	NH4+ and PO4 3-.
6	**(b)** 4	2, 8, 4.
7	**(a)** 48 km/h	Harmonic mean 2 x 40 x 60/100.
8	**(b)** 60 kg	Mass is constant.
9	**(d)** 120 J	Only vertical displacement counts.
10	**(a)** 0.00588 s	lambda/v.
11	**(a)** swell and may burst	Hypotonic environment.
12	**(b)** sclerenchyma	Dead and strong.
13	**(b)** fungus	Saprophyte.
14	**(a)** grafting	Joining two varieties.
15	**(a)** using public transport and saving energy	Reduces emissions.
16	**(d)** 20	72 x 5/18.
17	**(c)** is collected first	Evaporates first.
18	**(c)** copper	Metals have free electrons.
19	**(c)** 10.8 u	2 + 8.8.
20	**(b)** 3.14 m/s	2 pi r/T = 12.56/4.
21	**(c)** 8 N	Equals applied force.
22	**(d)** 45 rupees	9 units x 5.
23	**(d)** increases	Larger amplitude.
24	**(b)** 4	Half number.
25	**(a)** trachea	Moves mucus.
26	**(d)** bird	Aves.
27	**(b)** unisexual	Separate male and female.
28	**(c)** 700 W per square metre	70 per cent absorbed.
29	**(d)** 16.7 m/s	60 km/h x 5/18.
30	**(b)** 50 mL	0.2 x 250.
31	**(c)** 30 g	20 x 60/40.
32	**(a)** calcium	Z = 20.
33	**(c)** 40 m	v squared = 2as.
34	**(b)** 100 N	10/0.1.
35	**(c)** 50 N	MA = 4.
36	**(d)** half a wavelength	Half cycle.
37	**(a)** chloroplast	Plants only.
38	**(d)** cartilage and synovial fluid	Lubrication.
39	**(b)** more closely related than species in the same family but different genera	Closer relation.
40	**(a)** 46	Same number.
41	**(d)** 1000 tonnes	20 x 50.

42	**(b)** low pressure and high temperature	Real gases approach ideal behaviour.
43	**(b)** separating funnel	Immiscible.
44	**(c)** $\mathrm{Ca(OH)_2}$	Ca2+ and OH-.
45	**(c)** aluminium	2, 8, 3.
46	**(b)** uniform acceleration from rest	v proportional to t.
47	**(d)** 15 N	Balanced forces.
48	**(c)** 36 per cent	(5 - 3.2)/5.
49	**(d)** decibels	dB.
50	**(d)** mitosis	Mitosis.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** secondary xylem	Seasonal growth.
2	**(a)** monocot seed	One cotyledon.
3	**(d)** the 14th day of a 28-day cycle	Mid-cycle.
4	**(c)** run-off	Surface flow.
5	**(a)** 0.15	0.5 x 0.3.
6	**(a)** colloidal particles	Particles smaller than pores.
7	**(d)** O_2	Two shared pairs.
8	**(b)** neutrons	Mass number differs.
9	**(c)** 2r	Diameter.
10	**(c)** 50 N	Balanced.
11	**(d)** 5 m/s	v = sqrt(2KE/m).
12	**(d)** 343 m/s	f lambda.
13	**(c)** protein synthesis	Ribosomes make proteins.
14	**(d)** collenchyma	Corner thickening.
15	**(a)** chordate	Chordata.
16	**(b)** vegetative reproduction	New plantlets.
17	**(d)** converting N2 into nitrogen oxides	High energy.
18	**(c)** 0.2	2 x 0.5 x 0.2.
19	**(c)** 470 g	Solute 30 g.
20	**(a)** 12 g	9 x 4/3.
21	**(c)** 2, 8	Loses 2.
22	**(a)** 2.5 m/s squared	72 km/h = 20 m/s; a = 20/8.
23	**(d)** 4 N	a = 2.
24	**(a)** 500 W	15000/30.
25	**(c)** becomes higher	Higher frequency.
26	**(b)** isotonic	Equal concentration.
27	**(b)** bone	Rigid.
28	**(a)** platypus	Monotreme.
29	**(d)** cross-pollination (or between flowers)	Different flowers.
30	**(b)** uses less electricity	Efficiency.
31	**(a)** defines its limits and develops a better model	Model improvement.
32	**(b)** crystallisation	Cool saturated solution.
33	**(c)** Fe_2O_3	Fe3+ and O2-.
34	**(a)** carbon (2, 4)	4 for C; others 2, 1, 1.
35	**(a)** 5 m/s squared	20/4.
36	**(b)** 20 N	F = change in momentum / time = 1/0.05.
37	**(c)** 3	L/E.
38	**(b)** time period	T.
39	**(b)** detoxification	Detox.
40	**(b)** cartilage	Flexible pads.
41	**(a)** kingdom	Highest rank.
42	**(b)** pollen tube	The pollen tube carries the male gametes to the ovule.

43	**(b)** 80 per cent	Fraction x 100.
44	**(a)** 12.6 g	Average.
45	**(d)** colloid or suspension	Scattering.
46	**(a)** N_2	Three pairs.
47	**(b)** carbon dating	Radioactive dating.
48	**(b)** pi r	Half the circumference.
49	**(d)** 2 m/s squared	200/100.
50	**(a)** 2 m/s	sqrt(2 x 10 x 0.2).

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